

SHRUTI SALVI
Data Engineer & Analyst

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WORK EXPERIENCE

Metropolitan Transportation Authority (MTA – Metro-North Railroad)

New York, NY

Senior Data Analyst – System Safety (Full-time – Data Engineering & Analytics)

August 2025 – Present

- Analyzing and preparing monthly and annual safety data reports, trend reports, and statistical key data points to internal and external stakeholders in order to support important decision-making and risk mitigation.
- Creating insightful dashboards in Power BI and Tableau that enable data-driven decision-making and safety strategy optimization across the system safety department.
- Reviewing employee, customer, and contractor incident data for accuracy and quality assurance, and providing direction to employees to ensure compliance with corporate and federal reporting requirements.
- Performing incident record entry, validation, reconciliation, and database management using Cority and MS Access for analysis, auditing, and reporting.
- Building applications using Power Apps, writing Python scripts for data collection and cleaning, automating tasks by creating Power Automate flows, collecting and organizing data using MS Excel, and writing SQL queries in MS Access to retrieve data efficiently and accurately.
- Performing safety outreach efforts such as community events, tracks outreach, grade crossing, and gap outreach to support the safe operation of the railroad and create safety awareness across the transit system.

Metropolitan Transportation Authority (MTA – New York City Transit)

New York, NY

Information Technology Support Aide (Internship – Data Engineering & Analytics)

May 2024 – August 2025

- Engineered an MS Access database with SHA-256 authentication and VBA, adopted as a NYCT benchmark in 15+ projects worth \$50M each, boosting Azure Data Lake integration and report efficiency by 40%.
- Built a SharePoint document management system, migrating 35K legacy files and enhancing accessibility by 70%. Created a Power Apps leave request app integrated with Power Automate, reducing processing time by 100%.
- Orchestrating an ETL pipeline prototype in Google Colab using Python, SQL Alchemy, PySpark, and Azure Data Factory (ADF), targeting a migration time reduction from 3 days to 1 while adhering to data governance standards.
- Developing Power BI dashboards with filter indexing and optimized search, leveraging Azure Data Lake and Hexagon EAM software for faster data retrieval across various subway stations and 20+ device types, managing 4.4 million data points.
- Orchestrating the migration of legacy database systems (Access, Excel, and files) across various subway stations, focusing on configuration management and equipment asset data, ensuring seamless integration and transformation of historic and ongoing data into a centralized system using the EAM Flex SQL and Python environment for automation.
- Developing and implementing migration scripts to transfer asset data from legacy systems to a centralized EAM platform, ensuring seamless integration and accurate mapping of equipment and configuration data.
- Analyzed data from ridership metrics and 20+ sources to deliver insights on MTA performance & rider behavior. Built Tableau dashboards that improved operational efficiency by 80% in one month. Identified improvement areas in passenger retention & service optimization, collaborating with 10+ teams. Trained 40+ stakeholders on data flows using DOMO & BigQuery.

UNIQUE System Skills India Pvt. Ltd.

Pune, India

Programmer Analyst (Full-time – Data Engineering & Analytics)

August 2022 – July 2023

- Engineered scalable data pipelines using Apache Spark and Kafka, facilitating real-time data processing that improved analytics capabilities and reduced latency by 40%.
- Designed and implemented advanced algorithms in Python for data extraction and transformation, optimizing database interactions across MySQL and PostgreSQL environments.
- Enhanced data integrity and accuracy through the development of robust ETL processes in Informatica, achieving a 50% increase in data reliability.
- Resolved a 9-month recurring data mapping issue, reconciling 91% of discrepancies in health plan costs for 8K clients.
- Analyzed data from 159 sources to uncover trends in healthcare research & examined 5.9K patient records to identify patterns in insurance claims, reducing overhead costs by 70%. Sourced & processed 200GB of clinical trial data in compliance with GCP regulatory standards.

- Spearheaded the creation of dynamic data visualization dashboards with Looker and Google Data Studio, empowering stakeholders to make data-driven decisions that increased operational efficiency by 25%.
- Utilized R and Scala to develop predictive models that enhanced inventory management, resulting in a 30% reduction in stock shortages.
- Automated deployment workflows with GitHub Actions and Docker, streamlining development cycles and improving deployment frequency by 60%.
- Designed predictive models using Python, R, and SAS to analyze client behavior, improving Q1 retention rates by 74%.
- Leveraged data from Salesforce CRM and web analytics to identify 10+ new service opportunities, generating \$300K in annual revenue.
- Created dashboards to track sales, marketing effectiveness, and 20+ key metrics, increasing executive visibility by 48%.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

Master of Science (MS) in Information Systems | Granted with a partial annual scholarship for both years | GPA: 3.91/4.0

New York, NY
May 2025

Coursework: Python Programming | Database Management Systems (DBMS) & Data Structures | Data Engineering | Data Analytics | Database Programming | Information Systems Design & Development | Business Analytics | Big Data & Information Systems | Mathematics & Statistics | Cloud Computing | Java Programming | Data Science & Data Visualization | Data Mining & Machine Learning | Visual Design for Technology | User Experience Design | Information Security & Controls | AI & Ethics | Research Project

Activities and Societies: Pace Data Science Club, Pace Women in Tech, Pace Indian Student Association, International Buddy Program.

MIT ADT University

Bachelor’s Degree

Pune, India

TECHNICAL SKILLS & CERTIFICATIONS

Programming Languages: Java, JavaScript, R, Scala, Python-Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, PyTorch.
Data Engineering Tools: Apache Spark, Apache Flink, Apache Kafka, Apache Airflow, Apache Iceberg, Informatica, SSIS, SSRS, DBT.
Data Analysis & Visualization Tools: Tableau, Power BI, MATLAB, Google Analytics, Google Data Studio, RStudio, Looker.
Database Management Tools: SQL, MySQL, PostgreSQL, Microsoft SQL Server, Snowflake, Cassandra, Hadoop, Oracle, MongoDB.
Operating Systems: MacOS, Windows, Linux.
CI/CD Tools: GitHub, GitHub Actions, GitLab, Helm, Docker, Terraform, Kubernetes.
Cloud Computing Platforms: Amazon Web Services (AWS), Genesys Cloud, Microsoft Azure, Google Cloud Platform (GCP).
Certifications: Python Programming by Google (2022), Data Analytics by IBM (2022), Business Analytics by University of Colorado Boulder (2022), INSPIRE Program by Pace University (2023), Azure Data Engineer Associate by Microsoft (2024).
Languages & Other Skills: English, French & Microsoft Office Suite, Adobe Acrobat, Adobe Creative Cloud.

TECHNICAL PROJECTS

Cardiac Care Patient Data Analysis using Python

January 2024 – May 2024

- Collaborated in a team of five to analyze heart disease patients' dataset, assessing co-morbidities & prescription histories.
- Visualized the correlation between features & severity of heart disease using Plotly to customize long-term care plans.

Real-Time Patient Monitoring and Alert System using Apache Flink & Kafka

January 2023 – July 2023

- Led a team of four to design a patient monitoring system with Apache Flink, Kafka, and Snowflake to process IoT sensor data in real time.
- Deployed on Kubernetes with Airflow and automating CI/CD via GitHub Actions and Terraform reducing response time by 47%.

Comprehensive Data Analytics System for Enhancing Hospital Operational Efficiency

January 2023 – July 2023

- Led a team of four to design and deploy a data analytics solution leveraging Power BI, SQL Server, and Snowflake to analyze hospital operations, including patient flow, staff productivity, and treatment times.
- Utilized Python for data preprocessing and statistical analysis, with Apache Airflow for orchestrating ETL processes. Delivered actionable insights that led to a 20% improvement in operational efficiency and resource optimization.

Real-Time Credit Scoring System for Loan Approval using Apache Flink & Kafka

February 2022 – June 2022

- Collaborated in a team of six to build a real-time credit scoring system using Apache Flink, Kafka, and Snowflake to analyze transaction data and assess creditworthiness, enabling automated loan approvals.
- Deployed on Kubernetes with Airflow and automating CI/CD via GitHub Actions and Terraform, reducing loan approval and processing time by 56%.

Real-Time Fraud Detection Analytics for Financial Transactions using Python & Power BI

February 2022 – June 2022

- Collaborated in a team of six to conduct in-depth analysis of transaction data using SQL Server and Snowflake to uncover fraud patterns. Utilized Python for data cleaning, trend analysis, and feature extraction.
- Developed Power BI dashboards to visualize real-time insights, empowering financial teams to detect and prevent fraudulent activity, leading to a 25% decrease in fraud incidents.

HONORS & AWARDS

Career Services Award (Awarded by Pace University Career Services)

April 2025

- Honored for outstanding career development, demonstrating proactive engagement in professional growth initiatives, including mentorship, career coaching, and skill-building programs.
- Successfully leveraged these resources to secure a high-impact internship and advance career prospects in the data engineering field.

NBLK Consulting Hackathon – 1st Place (Organized by Pace University)

March 2025

- Led a team of 4 in developing AI hiring platform optimizing recruiting & screening process for small businesses, integrating NLP for intuitive chat-based assessments & presenting solution to startup founder, securing 1st place.
- Integrated quantum security to prevent unauthorized access to candidate/employer data, ensuring adherence to latest NIST framework standards.